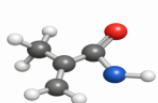


1 Inputs and dependencies



Molecule



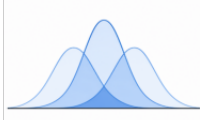
geometry

Charge



closed shell

Basis set



STO-3G, 6-31G(d)

PyQInt

$$\iiint \phi_i(r_1) \frac{1}{r_{12}} \phi_j(r_2)$$

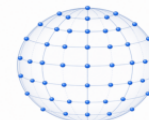
analytic integrals

NumPy



arrays

PyLebedev



angular rules

2 Core PyDFT construction



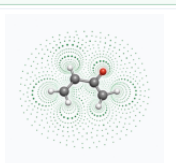
Basis amplitudes



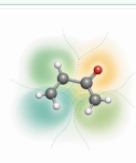
$$S_{\mu\nu}, T_{\mu\nu}, V_{\mu\nu}$$



Becke grid $\{\mathbf{r}_i, w_i\}$



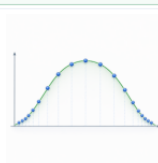
Fuzzy weights $w_A(\mathbf{r})$



Lebedev $\int d\Omega$



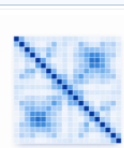
Radial grid r_k



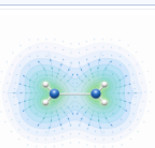
3 Kohn-Sham iteration



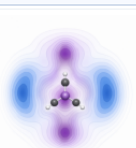
Density matrix $P_{\mu\nu}$



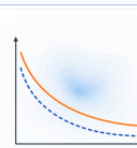
$$\rho(\mathbf{r})$$
$$\nabla\rho(\mathbf{r})$$



Hartree $v_H(\mathbf{r})$



XC $V_{xc}^{LDA/PBE}$



Build Fock $F = H + J + V_{xc}$



Solve KS $FC = SC\epsilon$



Update density $P = 2CC^T$



No

Converged?
 $|\Delta E|, |\Delta P| < \tau$

Yes

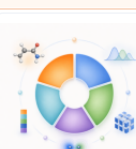
4 Exposed learning outputs



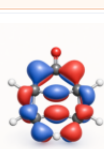
Matrices F, P, S



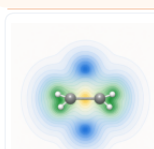
Energy terms E_i



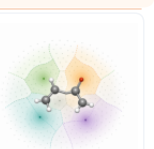
Orbitals ψ_i



Density maps $\rho(\mathbf{r})$



Becke cells $w_A(\mathbf{r})$



Spherical coeff. ρ_{klm}



Timing breakdown



Interactive exploration

